

Scalable Data Science
NPTEL 2026
Assignment-1

Probability

1. A data scientist launches four independent worker nodes in a distributed computing cluster. Each node processes between 1 and 6 GB of data per minute, depending on system load, with all rates equally likely. The total throughput is defined as the sum of the throughputs of the four nodes. What is the probability that the cluster achieves a throughput of at least 22 GB/min?

A $\frac{10}{1296}$
B $\frac{15}{1296}$
C $\frac{20}{1296}$
2. Let A and B be two events such that $P(A) = 0.7$ and $P(A \cap B) = 0.25$. What is $P(A \cap \bar{B})$?

(a) 0.45
(b) 0.25
(c) 0.70
3. There are 200 students enrolled in the Scalable Data Science course. The average number of hours spent per week by a student on course-related activities is 15 hours. How many students could be spending 75 hours or more per week on the course?

(a) At least 45
(b) At most 40
(c) Can't say
4. A machine learning algorithm is run independently n times, and each run succeeds with probability p . Let X be the number of successful runs. Assuming $p < \alpha < 1$, which of the following bounds the probability that at least an α -fraction of the runs are successful?

(a) $\frac{p(1-p)}{n(\alpha-p)}$
(b) $\frac{p(1-p)}{n(\alpha-p)^2}$
(c) Given information is incomplete

5. Consider the following game of dice: A player begins with a score of 0. The player rolls two dice at each turn t , and all of the numbers 1, 2, 3, 4, 5 or 6 are equally likely to come up. This is followed by taking product of two numbers on the two dice. The game ends when product at one turn becomes > 20 . On the contrary, if the product is < 20 , the product p gets added to their scores and a new turn t is awaited. The score of the player at the end of the game is thus $\sum_t p_t$ before the first turn in which they may get the product > 20 (where p_t indicates the product of dice at turn t). What is the probability of the player's score to be zero at the end of the game?
- (a) $\frac{1}{3}$
 (b) $\frac{1}{6}$
 (c) $\frac{1}{2}$
6. Let X be a random variable such that $\mathbb{E}[A] = 50$. Then, which of the following is correct?
- (a) $\mathbb{E}\left[\frac{1}{A-10}\right] \geq \frac{1}{40}$
 (b) $\mathbb{E}\left[\frac{1}{A-10}\right] \geq \frac{1}{10}$
 (c) $\mathbb{E}\left[\frac{1}{A-10}\right] < 40$
 (d) $\mathbb{E}\left[\frac{1}{A-10}\right] = 40$
7. A deck of 52 cards is shuffled followed by division into two halves of 26 cards each. From one of the halves, one card is drawn which turns out to be an ace. The ace is then put in the second half deck. This half is shuffled followed by drawing a card from it. What is the probability that the drawn card is an ace?
- (a) 0.094
 (b) 0.062
 (c) 0.071
8. Suppose there are 3 coins, out of which one is fair and rest two are biased. Let us pick one of the coins uniformly at random and flip it 3 times. The three coins comes with **heads** with a probability of 0.6, 0.1 and 0.5. What is the probability of the event HTT where H stands for heads and T for tails?
- (a) $\frac{302}{1000}$
 (b) $\frac{302}{3000}$
 (c) $\frac{177}{1000}$

Linear Algebra

1. For every real matrix $A \in \mathbb{R}^{m \times n}$, the matrices $A^T A$ and AA^T have exactly the same eigenvalues.
- (a) True
 (b) False

2. A reflection matrix in $\mathbb{R}^n$ is given by $R = I - 2vv^T$, where v is a unit vector. If the vector v itself is reflected by R , by what factor is it scaled?
 - (a) 1
 - (b) -1
 - (c) 2
 - (d) -2

3. Let M be a 4×4 matrix with real values, and let $x \in \mathbb{R}^4$ be a non-zero vector. What can we say about the vectors x , Mx , M^2x , M^3x , and M^4x ?
 - (a) Linearly independent
 - (b) Linearly dependent
 - (c) Linearly independent iff M is a symmetric matrix
 - (d) Cannot be determined.

4. Which of the following is equivalent to saying that B is a basis of a vector space V ?
 - (a) B is a minimal generating set of V .
 - (b) B is a maximal linearly independent subset of V .
 - (c) Every vector in V has a unique representation as a linear combination of vectors in B .
 - (d) All of the above.

5. Let $X \in \mathbb{R}^{n \times d}$ be a mean-centered data matrix. Suppose you perform PCA by computing the eigenvectors of the covariance matrix and select the top k eigenvectors to form the projection matrix $E \in \mathbb{R}^{d \times k}$. You obtain the lower-dimensional representation of the data as: $Y = XE$. Which of the following statements is correct regarding the ability to reconstruct X from Y .
 - (a) X can never be recovered from Y .
 - (b) X can always be recovered from Y .
 - (c) X can be exactly recovered from Y only if all d eigenvectors are used in E .
 - (d) X can be recovered even when only the top $k < d$ eigenvectors are used.

6. A function f is known to be uniformly continuous on a closed interval $[a, b]$. Which of the following conclusions can always be drawn?
 - (a) f has an interior maximum.
 - (b) f is bounded.
 - (c) f is differentiable.
 - (d) f' is continuous.

Machine Learning and Optimization

- 1 A dataset is generated from a nonlinear function with additive Gaussian noise. Polynomial regression models of degree 1, 2, and 10 are fitted using 20 training examples. Since higher-degree models contain all features of lower-degree models, which of the following statements is correct?
- (a) The 1st-order model has the lowest training error.
 - (b) The 2nd-order model has the lowest training error and the 10th-order model has the lowest test error.
 - (c) The 1st-order model has the highest training error and the 10th-order model has the lowest training error.
 - (d) None of the above.
- 2 A data scientist runs the k-means algorithm multiple times with different random initial centroids on the same unlabeled dataset. Which criterion should be used to select the best clustering among the different runs?
- (a) Use the labels of the data.
 - (b) Select the clustering from the last run.
 - (c) Select the clustering that minimizes the within-cluster sum of squared distances.
 - (d) Any clustering is equally good.
- 3 A machine learning engineer collects m training samples from a Gaussian distribution and computes the sample mean $\bar{X}$. Which of the following statements about the squared error

$$\sum_{i=1}^m (X_i - a)^2$$

is true?

- (a) It is minimized when $a = \bar{X}$.
 - (b) It is minimized when $a = \mu$.
 - (c) It has the same value for $a = \bar{X}$ and $a = \mu$.
 - (d) It is independent of a .
- 4 The closed-form solution for linear regression is given by:

$$\beta = (A^T A)^{-1} A^T Y.$$

When $A^T A$ is ill-conditioned due to multicollinearity, ridge regression adds an L_2 penalty:

$$\lambda \|\beta\|_2^2.$$

What is the resulting closed-form solution?

- (A) $\beta = (A^T A + \lambda I)^{-1} A^T Y$
- (B) $\beta = (A^T A)^{-1} A^T Y - \lambda I$

- (C) $\beta = \lambda(A^T A)^{-1} A^T Y$
- (D) $\beta = (A^T A - \lambda I)^{-1} A^T Y$

5 How can one solve a clustering algorithm stuck at a bad local optima?

- (a) Fix a seed value.
- (b) Use several random initialisations
- (c) Increase the number of data points
- (d) Both (a) and (b)

6 Among all cylinders having the same volume V , which of the following conditions is satisfied by the cylinder with the smallest surface area?

- (a) $h = 2r$
- (b) $h = \sqrt{\pi r^2}$
- (c) $h = \frac{V}{\pi r}$
- (d) There are infinitely many optimal solutions.

Answer Key

Probability

1. Correct answer (b).

$$\text{Total outcomes} = 6^4 = 1296.$$

For total throughput ≥ 22 , the possible sums are 22, 23, 24:

- 24: $(6, 6, 6, 6) \rightarrow 1$ way.
- 23: one 5 and three 6's $\rightarrow \binom{4}{1} = 4$ ways.
- 22: one 4 and three 6's $\rightarrow \binom{4}{1} = 4$ ways, or two 5's and two 6's $\rightarrow \binom{4}{2} = 6$ ways.

Thus, the number of favorable outcomes is

$$1 + 4 + 4 + 6 = 15.$$

Hence,

$$P(\text{throughput} \geq 22) = \frac{15}{1296}.$$

2. Correct answer (a). Since event A can be partitioned into two disjoint events,

$$A = (A \cap B) \cup (A \cap \overline{B}),$$

we have

$$P(A) = P(A \cap B) + P(A \cap \overline{B}).$$

Therefore,

$$\begin{aligned} P(A \cap \overline{B}) &= P(A) - P(A \cap B) \\ &= 0.7 - 0.25 \\ &= 0.45. \end{aligned}$$

3. Correct answer (b). By Markov's inequality,

$$P(X \geq 75) \leq \frac{E[X]}{75} = \frac{15}{75} = \frac{1}{5}.$$

Hence, the number of students spending at least 75 hours per week is at most

$$200 \times \frac{1}{5} = 40.$$

4. Correct answer (b). Since $E[X] = np$ and $\text{Var}(X) = np(1-p)$, by Chebyshev's inequality,

$$P(X \geq \alpha n) \leq P(|X - np| \geq n(\alpha - p)) \leq \frac{np(1-p)}{n^2(\alpha - p)^2} = \frac{p(1-p)}{n(\alpha - p)^2}.$$

5. Correct answer (b). $(4,6), (6,4), (5,5), (5,6), (6,5), (6,6) \therefore \frac{6}{36} = \frac{1}{6}$

6. Correct answer (a). Jensen's inequality gives

$$\mathbb{E} \left[\frac{1}{A - 10} \right] \geq \frac{1}{\mathbb{E}[A] - 10} = \frac{1}{40}.$$

7. Correct answer (a). 0.094

Using the linearity of expectation provides the fastest path to the solution:

- (a) Initial Condition: A standard deck has 52 cards with 4 aces. Drawing an ace on the first step is equivalent to removing 1 ace from the total deck, leaving a pool of 51 cards containing exactly 3 aces.
- (b) Expected Aces in Second Half: The second half consists of 26 cards drawn randomly from these remaining 51 cards. The expected number of aces it contains is:

$$E[\text{Aces initially in Half 2}] = 26 \times \frac{3}{51} = \frac{26}{17}$$

- (c) After the Transfer: Adding the first drawn ace increases the second half's size to 27 cards, and its expected ace count becomes:

$$E[\text{Total Aces in Half 2}] = \frac{26}{17} + 1 = \frac{43}{17}$$

- (d) Final Probability: Drawing a card from these 27 cards yields an ace with a probability of:

$$P(\text{Ace}) = \frac{E[\text{Total Aces in Half 2}]}{27} = \frac{\frac{43}{17}}{27} = \frac{43}{459} \approx 0.09368$$

Rounding to three decimal places gives **0.094**.

8. Correct answer (b). $\frac{302}{3000}$

By the Law of Total Probability, we compute the probability of the sequence HTT for each coin and average them based on their uniform selection probability ($\frac{1}{3}$):

- (a) Coin 1 ($P(H) = 0.6, P(T) = 0.4$):

$$0.6 \times 0.4 \times 0.4 = \frac{96}{1000}$$

- (b) Coin 2 ($P(H) = 0.1, P(T) = 0.9$):

$$0.1 \times 0.9 \times 0.9 = \frac{81}{1000}$$

- (c) Coin 3 ($P(H) = 0.5, P(T) = 0.5$):

$$0.5 \times 0.5 \times 0.5 = \frac{125}{1000}$$

Combining the paths weighted by the selection probability yields:

$$P(HTT) = \frac{1}{3} \left(\frac{96 + 81 + 125}{1000} \right) = \frac{1}{3} \times \frac{302}{1000} = \frac{302}{3000}$$

Linear Algebra

1. Correct answer (b). The matrices $A^T A$ and $A A^T$ have the same nonzero eigenvalues, but may have different numbers of zero eigenvalues since their dimensions can differ.
2. Correct answer (b). Since v is a unit vector, $v^T v = 1$, so

$$Rv = (I - 2vv^T)v = v - 2v(v^T v) = v - 2v = -v.$$

Hence, v is scaled by a factor of -1 , so the correct answer is -1 .

3. Correct answer (b). Linearly dependent.

The sequence gives us a set of exactly five vectors:

$$\{x, Mx, M^2x, M^3x, M^4x\}$$

These five vectors all reside within a 4-dimensional vector space ($\mathbb{R}^4$). By the fundamental properties of linear algebra, the maximum number of linearly independent vectors in any vector space is equal to its dimension.

Because the size of this set (5) exceeds the dimension of the space (4), these vectors are guaranteed to be linearly dependent regardless of the choice of matrix M or vector x .

4. Correct answer (d). All of the above.

A subset B is a basis of V if and only if it is both linearly independent and spans (generates) V . Each option provides an equivalent definition:

- (a) A generating set is minimal if and only if it is linearly independent.
 - (b) A linearly independent set is maximal if and only if it spans the entire space.
 - (c) Spanning ensures every vector has a representation, while linear independence ensures that this representation is completely unique.
5. Correct answer (c). PCA projects the data onto a lower-dimensional subspace:

$$Y = XE.$$

If only the top $k < d$ eigenvectors are used, information is discarded and the reconstruction

$$\hat{X} = YE^T$$

is only an approximation of X . Exact recovery is possible only when all d eigenvectors are included in E , making the transformation invertible.

6. Correct answer (b). Since a uniformly continuous function on the closed interval $[a, b]$ is continuous, and every continuous function on a compact set is bounded, f is bounded..

$$\text{Uniform continuity on } [a, b] \implies \text{continuity on } [a, b] \implies f \text{ is bounded.}$$

Machine Learning and Optimization

1. Correct answer (c). Since the polynomial models are nested, increasing the degree adds more features without removing any existing ones. Hence, the training MSE cannot increase with degree. Therefore, the 1st-order model has the highest training error, and the 10th-order model has the lowest training error.
2. Correct answer (c). Since k-means minimizes the within-cluster sum of squared distances (distortion), the best clustering among multiple runs is the one with the smallest value of

$$\sum_{i=1}^m \left\| x^{(i)} - \mu_{c^{(i)}} \right\|^2.$$

3. Correct answer (a). The sample mean

$$\bar{X} = \frac{1}{m} \sum_{i=1}^m X_i$$

minimizes the sum of squared errors

$$\sum_{i=1}^m (X_i - a)^2.$$

Therefore,

$$\sum_{i=1}^m (X_i - \bar{X})^2 \leq \sum_{i=1}^m (X_i - a)^2$$

for every real number a . Hence, the squared error is minimized when $a = \bar{X}$.

4. Correct answer (A). The ridge regression objective is:

$$\min_{\beta} \|A\beta - Y\|_2^2 + \lambda \|\beta\|_2^2.$$

Taking derivative w.r.t. β :

$$2A^T(A\beta - Y) + 2\lambda\beta = 0.$$

Rearranging:

$$A^T A\beta + \lambda\beta = A^T Y.$$

Factoring β :

$$(A^T A + \lambda I)\beta = A^T Y.$$

Thus:

$$\beta = (A^T A + \lambda I)^{-1} A^T Y.$$

5. Correct answer (b). Random initializations prevent bad local optima.
6. Correct answer (a). Using the volume constraint $V = \pi r^2 h$ and minimizing

$$S = 2\pi r^2 + 2\pi r h,$$

we obtain $h = 2r$.